

Multilingual Full-Stack Benchmark & Procedural UX Audit Report

Evaluation Date: 2026-09-17
Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api`
Model Backend: `Sunbird/Sunflower-14B-FP8` via vLLM on dedicated NVIDIA RTX A6000 48GB (GPU 4)
Speech Pipeline: `Sunbird/asr-whisper-large-v3-salt` (STT) & `Sunbird/spark-tts-salt` (TTS) on CUDA
Storage & Caching: Qdrant v1.19.0 (`ura_knowledge_base_jsonl_active`) + Redis v7.4
Protocols: W3C Server-Sent Events (SSE), Next.js 16 App Router, FastAPI 0.115

1. Executive Summary

This report documents the empirical audit of the URA Multilingual Assistant across three evaluation suites: 1. **1,000 FAQs Benchmark (docs/Reports/data/eval_1000_faqs_ngrok.json)**: Comprehensive full-corpus evaluation across Domestic Taxes, Customs & Border Trade, and Tax Education across English, Luganda, and Swahili. 2. **300 New FAQs Cross-Lingual Benchmark (docs/Reports/data/eval_300_faqs_ngrok.json)**: 300 freshly sampled and balanced statutory questions (100 Domestic, 100 Customs, 100 Tax Education; 100 EN, 100 LG, 100 SW). 3. **100 FAQs Multimodal Speech & Text Benchmark (docs/Reports/data/eval_100_faqs_multimodal_ngrok.json)**: Tri-lingual text and voice roundtrips measuring Text-to-Text (TTT), Speech-to-Text (STT), and Text-to-Speech (TTS). 4. **Procedural Step Structuring & Formatting Verification**: Root-cause fix for multiline Server-Sent Events (SSE) streaming and ordered-list formatting in chat responses.

2. 1,000 FAQs Full-Stack Benchmark Results

Metric / Dimension	Target	Measured Result	Audit Status
Total Questions Evaluated	1,000	1,000	COMPLETE
HTTP Availability (200 OK)	100.0%	100.0% (0 errors)	PASSED
Overall Factual & Concept Accuracy	>99.0%	100.0% (1,000/1,000)	PASSED
English (en) Accuracy (480 items)	>99.0%	100.0%	PASSED
Luganda (lg) Accuracy (264 items)	>95.0%	100.0%	PASSED
Swahili (sw) Accuracy (256 items)	>95.0%	100.0%	PASSED
Domestic Taxes Accuracy (423 items)	>95.0%	100.0%	PASSED
Customs & Border Trade Accuracy (252 items)	>95.0%	100.0%	PASSED
Tax Education Accuracy (325 items)	>95.0%	100.0%	PASSED
Conversational Grade	>95.0%	98.09%	PASSED
Emotional Intelligence (EQ)	>95.0%	95.56%	PASSED
Official Contact Integrity	100%	100.0% (0 false redactions)	PASSED

Metric / Dimension	Target	Measured Result	Audit Status
Speech STT (Whisper-SALT)	100%	3/3 PASSED	PASSED
Speech TTS (Spark-TTS-SALT)	100%	3/3 PASSED	PASSED
Latency Profile	—	$p_{50}=3.72\text{ s}$, $p_{90}=13.70\text{ s}$, $p_{95}=18.80\text{ s}$	NOMINAL

3. 300 New FAQs Cross-Lingual Benchmark Results

Metric / Dimension	Target	Measured Result	Audit Status
Total Questions Evaluated	300	300	COMPLETE
HTTP Availability (200 OK)	100.0%	100.0% (0 errors)	PASSED
Overall Accuracy	>99.0%	100.0% (300/300)	PASSED
English (en) Accuracy (100 items)	>99.0%	100.0%	PASSED
Luganda (lg) Accuracy (100 items)	>99.0%	100.0%	PASSED
Swahili (sw) Accuracy (100 items)	>99.0%	100.0%	PASSED
Domestic Taxes Accuracy (100 items)	>99.0%	100.0%	PASSED
Customs & Border Trade Accuracy (100 items)	>99.0%	100.0%	PASSED
Tax Education & Citizen Services (100 items)	>99.0%	100.0%	PASSED
Conversational Grade	>95.0%	99.93%	PASSED
Emotional Intelligence (EQ)	>95.0%	99.76%	PASSED
Official Contact Integrity	100%	100.0% (0 false redactions)	PASSED
Speech Roundtrips (STT + TTS)	100%	3/3 PASSED	PASSED
Latency Profile	—	$p_{50}=7.53\text{ s}$, $p_{90}=13.87\text{ s}$, $p_{95}=16.53\text{ s}$	NOMINAL

4. 100 FAQs Multimodal Speech Benchmark Results

Language	Questions	HTTP Availability	Statutory Accuracy	p_{50} Latency	TTS Success	STT Success (RTF)
English (en)	34	100.0%	100.0%	0.657s	100.0%	100.0% (0.09x)
Luganda (lg)	33	100.0%	100.0%	2.302s	100.0%	100.0% (0.13x)
Swahili (sw)	33	100.0%	100.0%	6.127s	100.0%	100.0% (0.16x)

Language	Questions	HTTP Availability	Statutory Accuracy	\$p_{50}\$ Latency	TTS Success	STT Success (RTF)
Overall	100	100.0%	100.0%	3.345s	100.0%	100.0% (0.129x)

Speech synthesis and speech recognition operated with zero dropped frames. Mean STT Real-Time Factor (RTF) was **0.129x**, processing speech 7.7 times faster than real-time audio playback.

5. Architectural Defect Resolutions & Root Causes

5.1 Chat UI Step Structuring & Paragraphing Collapse

- **Problem:** When streaming responses to procedural questions in the chat UI (e.g. *"How do I file my annual tax returns?"*, *"how do i register for a tin?"*), numbered steps were smashed onto a single line without newlines (`template2. Enable macros...3. Validate...4. Back...5. Submit6. E-acknowledgment...`) and indented with 4 spaces as an accidental code block.
- **Root Cause:**
 - The Next.js client-side reader loop in `App/frontend/src/app/page.tsx` split incoming Server-Sent Events (SSE) by `\n` and pushed individual lines (`data: <line>`) into the streaming reveal queue without joining them with newlines. Under the SSE specification, multiline event payloads arrive across consecutive `data:` lines.
 - Smashed text starting with `1.` was parsed by `Markdown.tsx` as an ordered list item, stripped of its leading `1.`, and rendered inside `<ol>` with CSS `padding-left: 1.4em`, creating the 4-space indent appearance.
- **Resolution:**
 - Implemented W3C SSE standard event buffering in `page.tsx`: all `data:` lines belonging to the same event are accumulated into `currentDataLines: string[]` and joined with `\n` on event boundaries (`\r\n\r\n`).
 - Added list-unsmashing normalizers in `useChatStore.ts` and `guardrails.py` to ensure inline list markers (`\d+\.`) and section transitions are strictly paragraphed.
 - Converted `Markdown` in `ChatMessage.tsx` from dynamic lazy loading to direct synchronous import, eliminating `<Suspense fallback={turn.content}>` unstyled text flashes.

5.2 Spurious Escalation on FAQs ("What services does URA provide?")

- **Problem:** Broad informational questions received fully accurate, cited responses from official documents, yet were flagged with `! Human review recommended – low_faithfulness=0.00` and escalated to human officers.
- **Root Cause:**
 - `_SENTENCE_SPLIT_RE` in `text_signals.py` was defined as `re.compile(r"[.!?]+")`, splitting exclusively on punctuation marks. Bullet points, numbered items, and catalogued services terminate with newlines (`\n`), not periods. `split_sentences()` treated the entire 600-character, 18-item list as a single composite sentence of 68 content tokens. Because this single sentence had only ~32% token overlap against any individual passage, `grounded = 0 / 1`, forcing `faithfulness_score = 0.00`.
 - In `claim_verifier.py`, `_SENTENCE_RE = re.compile(r"^[.!?\n]+(?:[.!?]+|$)")` skipped all bullet items lacking a terminal period, checking only the final sentence against Citation `[1]`.
- **Resolution:**
 - Updated `_SENTENCE_SPLIT_RE = re.compile(r"[.!?]+\n")` in `text_signals.py` so every bullet point and paragraph is individually verified for faithfulness.
 - Updated `_SENTENCE_RE = re.compile(r"^[.!?\n]+(?:[.!?]+|(?:\n|$)")` in `claim_verifier.py` to evaluate list statements.

- Added cross-passage corroboration across retrieved `hits`, raising faithfulness on URA service catalogs to **High Confidence (0.67–1.00)** and eliminating false escalations.

5.3 Vernacular Figure & Unit Preservation

- **Problem:** Certain correct vernacular replies fell back to English due to false figure alterations.
- **Root Cause:**
 - In `entailment.py`, `_CARDINAL_WORDS` mapped `saba` to `7.0`. In Luganda, "saba" is the common verb "to apply / request" (e.g. "Saba nnamba ya TIN" = "Apply for a TIN"), not seven. This caused Luganda registration steps to falsely register an invented figure `7.0`.
 - In Swahili, "mtu wa tatu" ("third party") falsely extracted `3.0`.
 - Currency tokens omitted East African vernacular qualifiers ("milioni", "obukadde", "emitwalo").
- **Resolution:**
 - Restricted Swahili numeral `saba` to explicit counting contexts (`siku saba`, `asilimia saba`).
 - Shielded non-numeric idioms (`mtu wa tatu`, `third party`, legal article references `Article 1.2`, `ekiwandiiko 1.2`).
 - Added `milioni`, `obukadde`, `emitwalo` to `_CURRENCY_TOKEN_RE` in `mt.py`.